

Talent Intelligence Platform: Core Interaction Architecture

Confidential & Proprietary

This whitepaper details the architectural sequences and asynchronous queuing subsystems that power the Taledge Dual-Track Talent Intelligence Ecosystem. In strict accordance with the product vision, the architecture supports two fundamental branching paths: **Track 1 (Placement & Career Success)** and **Track 2 (Competitive Exam Success)**.

1. GOAL-DRIVEN INITIALIZATION & TRACK ASSIGNMENT

The platform operates on a dynamic branching architecture. During onboarding, a critical branching point determines the execution environment. The Orchestration Engine hydrates a strict execution context based on the user's primary objective.

1.1 The Track 1 (Placement) Context Payload

For Placement Institutes and Recruiters, the payload aggregates historical capability data, Target Role Vector Embeddings, and Academic markers.

1.2 The Track 2 (Competitive Exam) Context Payload

For Exam Aspirants, the payload completely bypasses resume matrices and instead aggregates longitudinal preparation metrics, exam syllabus difficulty vectors, and target examination frameworks (e.g., Civil Services, GATE, GRE).

1.3 High-Availability State Management

During initialization, the payload is committed to a distributed in-memory datastore. This ensures that whether a candidate is engaged in a 15-minute mock interview or a 6-month competitive exam preparation cycle, the state is instantly retrievable.

2. DETERMINISTIC PERSONA ROUTING MATRIX

Depending on the assigned Track, the interaction is routed to highly specialized sub-networks within the generation cluster.

2.1 Track 1: The Technical & Behavioural Adversary

In the placement pathway, the engine operates as a hiring manager.

- **Execution Directives:** It cross-examines the candidate strictly on skills and projects mentioned in the ATS payload. It induces cognitive load by challenging architectural constraints.
- **Goal:** To evaluate professional readiness, communication clarity, and problem-solving depth for immediate workforce deployment.

2.2 Track 2: The Counselling-Led Diagnostic Engine

The competitive exam pathway is fundamentally different. It is tailored to long-duration, high-stakes environments. The engine operates as a domain expert counselor.

- **Execution Directives:** The interaction is exploratory and diagnostic. It bypasses technical correctness entirely and focuses on uncovering stress levels, emotional patterns, preparation habits, and coping mechanisms.
- **Goal:** To identify early behavioural risk patterns (declining motivation, inconsistency) and prevent student burnout before failure cycles occur.

3. THE BI-DIRECTIONAL STREAMING PIPELINE

For both tracks, the conversational loop utilizes an ultra-low latency, bi-directional asynchronous stream.

3.1 Auditory Ingestion & Pre-Processing

The local client captures audio chunks, feeding them into a Voice Activity Detection (VAD) buffer. Once a pause interval is detected, the native transcribing engine fires the finalized text payload. The raw text is passed through an edge-normalization function to strip filler words and extract core semantic intents.

3.2 Inference Generation & Chunked Synthesis

The backend streams the normalized history to the primary generation network. To eliminate waiting latency, the output stream is chunked by sentence delimiters. As the first sentence is generated, it is pushed via an event stream and instantly synthesized into natural, emotion-aware speech with human-like intonations.

4. INTEGRITY WRAPPERS & MICRO-EXPRESSION TRACKING

4.1 Proctoring & State Monitoring

A strict visibility listener ensures focus. In Track 1, context-switching triggers immediate cheating flags. In Track 2, context-switching is logged as a potential indicator of distraction and declining discipline.

4.2 Latency Variance Tracking

The system calculates the millisecond delta between the engine finishing its speech output and the candidate's initial vocalization. High variance maps

directly to guessing (Track 1) or elevated psychological stress/uncertainty (Track 2).

5. PRIVACY, COMPLIANCE, & ETHICAL OUTPUT

5.1 Strict Access Controls (Exam Institutes)

In Track 2, identified behavioural risk patterns (e.g., elevated stress or self-harm tendencies) are strictly non-clinical. Access to these indicators is highly controlled—visible *only* to the candidate and their competitive exam training institute. They are never shared with placement entities or recruiters, ensuring absolute ethical integrity.

5.2 Enterprise Cryptography

Every payload sent over the network is verified using a short-lived, cryptographically signed token. Data at rest is encrypted via AES-256 standards.

6. HIGH-THROUGHPUT POST-PROCESSING

Upon interaction termination, the transcript, latency metadata, and proctoring logs are compiled into a finalized **Assessment Ledger**. This ledger is placed into an asynchronous event pipeline, where the decoupled Scoring Engine consumes it to map the data points against the overarching DNLA framework and generate the comprehensive Success Potential Score.